

## Kryptos Thoughts

It's 2018 and the code of the fourth part of the "Kryptos" sculpture at CIA known as "K4" has been unsolved for more than 27 years. With the huge growth in computer speed since 1990, the development of massive online databases of digitised books, texts, and statistics, and clues released by the sculptor in 2010 and 2014, is a solution now more likely than ever?

On the other hand, there has been a marked decrease of interest in the sculpture. One of the founders of the Yahoo Kryptos mailing list, Elonka Dunin, hasn't updated her timeline much since [April 2006](#).

Perhaps [people are collectively smart](#) and they realise that traditional methods aren't going to be successful: there's been a mistake or the sculptor Sanborn has stepped outside the parameters or [constraints](#) suggested to him by his CIA advisor, Ed Scheidt, and inadvertently come up with something nobody is ever likely to break. Or they realise that now that two plaintext clues have been released with nobody any closer to a solution, the smartest course of action is just to wait for more clues.

Sanborn did [say](#) he'd be *"modifying systems and developing my own which would make it virtually impossible for [Scheidt] to decipher all of it"*, there are mistakes in the first three parts, and he has even hinted that a unique decipherment [might not be possible](#). Objectively speaking this is rather discouraging for would-be solvers – if he expects a CIA cryptography expert cannot solve it, why should anyone else be able to?

On the mildly positive side, Scheidt did say [in 2005](#) that he was "confident" the part four encipherment had been done correctly and Sanborn said in 2006 he was ["pretty sure" about part four](#).

I recommend some background reading of pages and articles written since the 2010 "BERLIN" clue was released. If you have never read anything about Kryptos before, then start with the [Schridde post](#).

### Kryptos web pages in the "Berlin Clock" era

- 25 April 2011. [Dean L. Wiley](#) posts a PHP script on his webpage which attracts a large number of possible keywords (and injection attacks). Many suggested keywords are there: DYAHR, HYDRA, LAYERTWO, KLEPSYDRA, PARASYSTOLE, DIGETAL, LUCID, SHADOW etc.
- 1 July 2011. [Crypto Crap](#) who also posts as "skintigh" on reddit writes a summary of their thoughts. There is some good discussion on whether the "mistakes" are deliberate or not.
- 23 July 2011. [Please Decipher Me](#) writes about a fractionated substitution method. Linking from [zetaboards](#), he wrote *"I do believe that if you have an unknown algorithm, a short plaintext, and only one message, that there has to be some hint as to what the algorithm is. If someone posted an unknown algorithm cipher challenge here with ~100 characters of ciphertext with no clue as to the algorithm, no one would even bother."* In the same thread, pineconedegg commented: *"It's very classically unsolved for a reason, I assume. ... Unknown algorithm cipher challenges are just terrible, as cool and mysterious as Kryptos is."*
- 20 March 2013. [Kryptos Fan](#) publishes Gillogly's responses to some interview questions. The last post on this blog is around the beginning of 2015.
- 27 April 2016. [Bauer, Link and Molle](#) publish a paper in Cryptologia entitled *"James Sanborn's Kryptos and the Matrix Encryption Conjecture"*. In July 2016 one of Bauer's friends [Klaus Schmeh](#) publishes an article in German on his blog about it.
- 18 January 2017. [Patrick Kellogg](#) posts a presentation focusing on Vigenere solutions. *"There has been a notable decrease of interest in the sculpture, and several websites have stopped updating progress and theories."*
- 3 and 17 March 2017. [Christian Schridde](#) publishes a two-part explanation of the methods used for Kryptos parts 1 to 3 and examines the Bauer et al paper.

### Commentary on these webpages

#### **Gillogly Interview**

He wrote: *"... we know that we don't know the system used to encrypt it: Scheidt has said it's his own invention and hasn't been seen in the world before."*

This seems to be based on a misunderstanding. This was corrected at a [October 2013 American Cryptogram Association dinner](#). *"Elonka jumped in to remind Ed that he once told her that Kryptos uses a system unknown to anyone on the planet. Ed stated he didn't recall that conversation and back-pedaled out of saying anything more."*

The closest Scheidt came to saying this publicly was *"the masking technique may not be known"* in [Wired in January 2005](#) or possibly *"[the code] is unique to Jim's design" in 1991*. The "own invention" quote may be a misinterpretation of the first statement.

*"One question on which I'd like to see more consideration or insight is the issue about the end of K2. The NSA team, David Stein and I all decrypted it as it stands: ID BY ROWS. Several years later Sanborn said that he'd made an error, leaving out an X in the plaintext, and it should actually have said X LAYER TWO. I believe Scheidt raised an eyebrow about this explanation in print, though I don't have his quote in front of me.*

*I feel that having it decrypt two ways producing by chance perfectly grammatical English (both versions obscure, of course) is extremely unlikely. I think it's more likely that Scheidt put that secondary meaning in by a judicious selection of the keyword."*

I haven't seen this quote by Scheidt - does anyone know what he means? Is something which was mentioned only in the Kryptos Yahoo group and not outside it?

## Crypto Crap

The commentary on the "extra L" here is: *That extra L was added for "aesthetic reasons."* (read that somewhere, can't find a link though... did I imagine it?) Either [kerning](#) was adjusted to make room for the L, or mistakes in kerning were made requiring this L. Sanborn has left this L off of the models of Kryptos he sells and says it is not important for solving Kryptos, so that suggests it was a kerning mistake.

This again seems to be a misunderstanding possibly deriving from the [English Wikipedia article \(pre 16 Feb 2018\)](#) which says *"One of the lines of the Vigenere tableau has an extra character (L), which Sanborn has indicated was accidental."* The source given is a [November 2014 Wired article](#) which doesn't contain any similar statement. The ["aesthetic reasons"](#) quote actually refers to the LAYERTWO/IDBYROWS part.

The bit about *"Sanborn ... says it is not important"* seems to be just wrong: he didn't say that, that's excessive paraphrasing and/or overinterpretation.

Another article from The Magazine by Mark Siegal summarises a 2005 Kryptos dinner meeting: *"That letter is missing from the replica, so it was probably a mistake made while cutting out all the sculpture's letters."*

[Chris Hanson](#) had a similar conclusion from the same 2005 dinner: *"DYAHR is important to the puzzle (since Sanborn remembered it) ergo the extra L is not for the same reason."*

[Doug Gwyn](#) notes stated: *"The extra L on the tableau seems even more likely to be simply a production error, since Sanborn's models we saw at Cafe Asia didn't include them yet he said that they were supposed to have everything needed to solve."*

On the other hand there are some other facts that make the extra L seem useful which are discussed in the next section.

In 2005 the sculptor commented on CNN that there was ["one clue"](#) on the Kryptos version which wasn't on Antipodes (a near replica of Kryptos at the Hirshhorn Museum in Washington DC).

There are only two strict textual differences between the sculptures - on Kryptos, the "UUND" part of the word "UNDERGRUUND" encrypts to "RTBJ" while on Antipodes, the spelling has been corrected, as in the original worksheet, to "OUND" and thus encrypts to "ETBJ". The second difference is the extra "L" (spelling "HILL" on the right hand edge) is present only on Kryptos.

[Frank Corr of the ACA](#) agrees with CryptoCrap that the misspelling in Kryptos part one is due to an encoding error on the [original worksheet](#). i.e. PALIMPSEST as PALIMPCEST.

## Bauer, Link and Molle paper and Schridde post

This paper and post examine the idea of a "Hill Cipher" or matrix encryption being used in K4. Before plaintext was released for K4, Gillogly and the NSA team guessed the probable methods for this section were combined transposition and polyalphabetic substitution, autokey, or running key.

Elonka's timeline mentions that Roger Anderson noticed the extra "L" on the sculpture in [October 2003](#). The NSA memorandum of [March 1993](#) had already noted this.

Following this [Keith Edkins looked at the 2x2 Hill Cipher in February 2004](#). He examined the Kryptos alphabets plus the standard alphabet, in all shifts, for the first 96 and last 96 characters of the ciphertext, using all 157,248 invertible 2x2 key matrices.

[Krazy Kryptos](#) had looked at the 2x2 Hill Cipher in December 2010.

After the clues of 2010 and 2014 were released, Bauer et al published a 2016 paper looking at order 2 and 3 matrix encryption for K4.

In his [YouTube talk of 24 April 2013](#) around 25:00 and [International Spy Museum talk of 18 July 2017](#) around 51:45 Bauer also mentioned this theory.

There are several serious problems with the Bauer et al paper. As Bauer is the editor of *Cryptologia* this may indicate that the paper did not go through a stringent refereeing process. It may also have been intended as more of an educational paper.

- The authors do not consider the different alignments. For instance, Edkins looked at the two possible alignments for the 2x2 case; but the paper only considers one each for the 2x2 and 3x3 cases, out of a possible five. This was always going to be a problem with a prime number of cipher text characters. Then again, [Ed Scheidt in Wired](#) did say once there were 98 characters left, which is divisible by 2 and 7. Further, the paper fails to recognize that in the 2x2 case where "BE" is aligned with "NY" (characters 64-65) this requires omitting the initial "O" of the ciphertext, but in the 3x3 case where "BER" is aligned with "NYP" this requires omitting the final "R" of the ciphertext. The attempted decryptions are considering different offsets of the ciphertext and the authors seem unaware of it. They missed what Edkins had noticed in 2004.
- Using the known plaintext, it's quite possible (for a given alphabet mapping) to recover the key matrix row by row for the 2x2 and 3x3 cases ([Bauer and Millward](#)). Instead, for the 3x3 case, the paper contains text about building a FPGA and how the search took many hours. Recovering the key matrix row by row (i.e.  $3 \times 26^3$  possibilities instead of  $26^9$  possibilities - [James Lyons has also mentioned this possibility](#)) reduces the time for this search for each alphabet mapping to a fraction of a millisecond. Using this method, it is even feasible to test all the sculpture alphabets for the 4x4 case in a short time. We can also check the Affine Hill Cipher concept (a matrix equation of the form  $AX+B$ ) for the sculpture alphabets with 2x2 and 3x3 matrices. This eliminates one of the four Bauer et al suggestions in the paper.
- The paper [source code](#) is not commented and does not have any documentation. I could not figure out how the program worked. It is not useful to anyone seeking to improve it or generalize it.

Bauer's "Unsolved!" book, where he describes the same results, is available for about \$US10 (Kindle edition). Taylor and Francis want \$US50 to download a PDF of the *Cryptologia* paper which is the same as Sanborn wants since 2014 as a "correspondence fee". Bauer doesn't provide a PDF of the paper on his website. On 12 June 2017 the article had had 92 views since it was published on 27 April 2016 (through the Taylor and Francis website, as opposed to sci-hub).

This retards the progress of science and blocks access to interested people outside academia. I'd be willing to pay about \$US0.50 for the PDF if I didn't have it. I can see the parallels with the music industry circa 1999.

Historically, in the two Hill papers and the Friedman books (mentioned below) people tend to work modulo 26 i.e. mappings 0 to 25 - numbering 1 to 26 hasn't really been considered in the literature. There's a [second Hill paper](#) from 1931 containing a different alphabet which was not considered – MJDXAHOU CZQETYFWGIVSKPLRNB.

Three reasons were given as hints that matrix encryption was used for K4. It seems quite plausible that "HILL" down the right-hand edge of the sculpture is indeed referring to the Hill Cipher. The other two suggestions about the coordinates of the errors in K1-K3 having a role in decrypting K4 and "BERLIN" being aligned with the cipher are more tenuous and seem like wishful thinking. Certainly, Sanborn stated in Wired [in April 2006](#): "... clues to the last section, which has only 97 letters, are contained in previously deciphered parts. Therefore, getting those first three sections correct is crucial." However, Bauer is in a good position to know more than others about the possible methods, being editor of *Cryptologia* and having worked in the area of unsolved ciphers for some time.

Also, Jim Gillogly already stated in September 1999: "*I was told by Sanborn in a phone conversation after I did my part of the solution that he'd given the CIA a big hint a year or two ago*". Thus, hints are given to some people and not others.

There are some other good reasons, not mentioned in the paper, to believe that matrix encryption is likely, apart from the Antipodes comparison mentioned above.

- In 2007, Sanborn stated to PBS that “matrix codes (bar codes)” were his favourite type of encryption, being more “art-like”.
- In Wired in 2005 he stated he “used some matrix codes Ed gave me”.
- The “Atomic Time” book of 2003 quotes Sanborn: “a fairly extensive text [was encoded] into a matrix system”.
- At a 2005 dinner, Sanborn mentioned that Kryptos is now featured in modern high-school math texts, in a section on matrices. [Chris Hanson] mentioned that Kryptos didn't use mathematical matrices, and he refuted “but it does”.

In rough terms, there are  $26!$  or about  $4 \times 10^{26}$  possible mappings for the alphabet, and for the 2x2 case, about 1 in a billion mappings have a corresponding 2x2 matrix which can produce “BERLINCLOCK”, for the 3x3 case, about 1 in a thousand, and for the 4x4 case, there might be about a million such 4x4 matrices for each mapping (although for some mappings there are none).

After a large scale search, I am confident a 2x2 matrix, with the same mapping for enciphering and deciphering text, was not used; at least, not if the rest of the plaintext is in plain English. The search looked at all possible 2x2 invertible matrices and for each matrix derived all possible alphabet mappings that would encipher “BERLINCLOCK” to “NYPVTMTZFPK”. The “leftover” letters not used in the known plaintext or ciphertext are A, D, G, H, J, Q, S, W, U and X.

Briefly, by adding mappings for H, A, and S, and possibly U depending on the alignment, then scoring the possible decipherments (while permuting the leftover letters) using modified [quadgram frequencies from Lyons](#) I found that no possible decipherment made sense in English.

As one example, it would have been great if, say “NFB” decoded to “THE” for “THE BERLIN CLOCK” and “SOLIFB” decoded to “ANDTHE”, since the only words which occur in all three of the previous sections of Kryptos are “AND” and “THE”. There are many ways this could happen but in every case the rest of the text is just gibberish.

The Schridde post follows on from the Bauer paper. In terms of the Hill cipher approach, he helpfully explains how instead of working with just  $n$  characters at once,  $n \times n$  characters can be encrypted. The cipher can then possibly be solved by choosing a keyword for processing into an alphabet mapping.

Using all the [ACA word lists](#) in one big file, searching about 1.9 million possible mappings derived directly from the words takes about 10 minutes for exploring the 3x3 cases on my laptop.

Extending that, Friedman and Callimahos in “Military Cryptanalytics Part I” suggest many ways that a keyword can be transformed into an alphabet mapping. To quote them:

*Let us examine several types of mixed sequences, using the key word HYDRAULIC as an example. The ordinary keyword-mixed sequence produced from this key word is;*

*(1) HYDRAULICBEFGJMKHQPQSTVWXZ*

*The two principal transposition-mixed types based on this key word are derived from the diagram*

HYDRAULIC

BEFGJMKMNG

PQSTVWXZ

*and read:*

*(2) Simple columnar*

HBPYEQDFSRGTAJVUKWLMXINZCO and

(3) Numerically-keyed columnar

AJVCODFSHBPINZLMXRGTTUKWYEQ

Other types may arise from various types of route transpositions such as the following, using the foregoing diagram:

(4) Alternate vertical

HBPQEYDFSTGRAJVWKULMXZNICO

(5) Alternate diagonal

HYBPEDRFQSGAUJTVKLIMWXNCOZ

(6) Simple diagonal

PBQHESYFTDGVRJWAKXUMZLNIOC

(7) Alternate horizontal

HYDRAULICONMKJGFEBPQSTVWXZ

(8) Spiral counterclockwise

OCILUARDYHBPQSTVWXZNMKJGFE

Still other types are possible from the foregoing diagram which do not follow a simple, clear-cut route, such as the following:

(9) HYEBPQSTGFDRAUKJVWXZNMILCO

(10) CPIOQBLNSEHUMZTFYAKXVGDRJW

They then go on to discuss the inverse of these mappings. The example mapping for the Hill Cipher later in the book (page 185) is also based on the keyword HYDRAULIC (number 3 above).

The named methods above were taken from a previous handbook by Parker Hitt - Chapter V of the 1916 [Manual for the Solution of Military Ciphers](#). The book by Friedman and Callimahos was published by Aegean Park Press in the 1980s so was public knowledge when Kryptos was dedicated in 1990.

It is interesting that HYDRAULIC here begins with "HYDRA", an anagram of "DYAHR" from the sculpture cipher text. "HYDRA" also referred to a CIA computer of the 1960s and 1970s for keeping tabs on Americans, explained in a [Church Committee report](#). What is meant by the letters on the sculpture: "DYAHR", "NDYAHR", "ENDYAHR" or "YAR" remains a mystery. At [Realm of Twelve](#) the other theories are that it refers to DAY and HR, and is related to the Hydra of [Camp X](#).

I had thought that perhaps "YAR" could have been derived from a diagonal type method like the above for the keyword KRYPTOS after putting it into a width 7 matrix.

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KRYPTOS KOPRSTY
ABCDEFGH ABCDEFG
HIJLMNQ HIJLMNQ
UVWXYZ  UVWXYZ
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For example **PYARSEKODITCHNBGMWFLVJUQZX**, **XZVQWUMJNLHFIGDBECKYARTSPO** or **QZNGXMFSWLEOVJDTUICPHBYARK**. But again, this idea didn't lead anywhere.

There's also the small possibility that a larger matrix has been used, but [there are many possibilities](#), so the search space needs to be compressed somehow. Naturally the possibilities become more outlandish.

For example, we could:

- combine the 25 letters of known keywords PALIMSPET, ABSCISSA and KRYPTOS into a 5x5 matrix using one of those methods
- anagram those to PSITSSIMPLEASABC
- add [PARASYSTOLE](#) to get 36 letters for a 6x6 matrix
- use a keyword with all different letters (e.g. RAY, HYDRA, KRYPTOS, DIGETAL, SCHEIDT, LAYERTWO, KLEPSYDRA) and produce a Latin square for the secret matrix based on that keyword.
- Create a matrix using just the values 0 and 1; or just 0, 1, and -1; or just 0, 1, -1 and 13. Even with just the values 0, 1 and -1 in a 2x2 key matrix there are many alphabets which can give the known plaintext with the correct choice of alphabet mapping.

We could possibly interpret the “YAR” as a reference to “ARY”, for example binary or 3-ary values in a matrix. It's interesting to ask: what is the minimum order binary matrix we'd have to choose with the sculpture alphabets as mappings to get “BERLINCLOCK” out?

If we take the alphabet ABC...KRYPTOS from the sculpture and number the letters appropriately we can obtain, for example, the following mapping.

|   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| A | B | C | D | E | F | G | H | I | J | L  | M  | N  | Q  | U  | V  | W  | X  | Z  | K  | R  | Y  | P  | T  | O  | S  |
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

[illegible]

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| ? | O | B | K | R | U | O |
| X | O | G | H | U | L | B |
| S | O | L | I | F | B | B |
| W | F | L | R | V | Q | Q |
| P | R | N | G | K | S | S |
| O | T | W | T | Q | S | J |
| Q | S | S | E | K | Z | Z |
| W | A | T | J | K | L | U |
| D | I | A | W | I | N | F |
| B | N | Y | P | V | T | T |
| M | Z | F | P | K | W | G |
| D | K | Z | X | T | J | C |
| D | I | G | K | U | H | U |
| A | U | E | K | C | A | R |

To keep the search time for finding such binary matrices reasonable we'd have to restrict ourselves to the sculpture alphabets or alphabets derived from the KRYPTOS keyword.

At higher orders even calculating the determinant to check if a matrix is invertible modulo 26 is quite time-consuming.

We could also use a different alphabet for the enciphering and deciphering mappings. This possibility is mentioned in Sinkov's "Elementary Cryptanalysis" book but not explicitly in the Friedman books or the Hill papers.

As a contrived example, say we took the words *afsnauwt* and *johdannossa* (Dutch and Finnish, from the [ACA word lists](#)) giving the alphabets AFSNUWTBCDEGHIJKLMNOPQRVXYZ and JOHDANSBCEFGIKLMPQRTUVWXYZ.

Enciphering mapping for plaintext

|   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| J | O | H | D | A | N | S | B | C | E | F  | G  | I  | K  | L  | M  | P  | Q  | R  | T  | U  | V  | W  | X  | Y  | Z  |
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

Deciphering mapping for ciphertext

|   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| A | F | S | N | U | W | T | B | C | D | E  | G  | H  | I  | J  | K  | L  | M  | O  | P  | Q  | R  | V  | X  | Y  | Z  |
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

We then encipher the plaintext "ERLI" as the matrix 9, 18, 14, 12 (by columns) and decipher the ciphertext "YPVT" as 24, 19, 22, 6 and use our secret matrix 17, 3, 5, 4 as follows to decipher the "YPVT" part of the ciphertext.

$$\begin{pmatrix} 17 & 5 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 24 & 22 \\ 19 & 6 \end{pmatrix} = \begin{pmatrix} 9 & 14 \\ 18 & 12 \end{pmatrix}$$

Unfortunately, the context of the decrypted ciphertext is then ... MAKABERLINCLOCKFBUEU ... which is no use.

The other sections of Kryptos were enciphered using [pen and grid paper](#) and the larger the matrix, the more difficult the encipherment is and the greater the possibility of error. This theme is heavily emphasised in the Sinkov book in the chapter on polyalphabetic encryption.

With cryptanalysis of the Hill Cipher method, it's critical that there are no mistakes in the encipherment of the known plaintext ("NYPVTTMZFPK" from "BERLINCLOCK") otherwise we really are wasting our time.

We can look at these Hill cipher decryption attempts in information theoretic terms.

There are  $26!$  or about  $4 \cdot 10^{26}$  possible permutations of the English alphabet. This corresponds to about  $\log_2(26!) = 88.4$  key bits.

There are  $26^{11}$  possibilities for the 11 known plaintext letters – about  $4 \cdot 10^{15}$  which is 51.7 bits.

If we were to use a 7x7 binary matrix, there are 163,849,992,929,280 invertible binary 7x7 matrices, which corresponds to about 47.2 bits. If we were to pick one of 27 possible alphabets (the sculpture alphabets plus the standard alphabet) that corresponds to about 4.8 bits. Essentially the number of possibilities is about  $47.2 + 4.8 = 52.0$  bits which is about the same amount of information in the 11 known plaintext letters. On average we'd expect to find about one "solution" (i.e. mapping BERLINCLOCK to NYPVTTMZFPK) in the keyspace.

With a 3x3 Hill Cipher there are 1,634,038,189,056 invertible matrices and for a 4x4 Hill Cipher 12,303,585,972,327,392,870,400 – 40.6 and 73.4 bits respectively. Even with the reduction of the keyspace due to the known plaintext there is an enormous number of possibilities.

David Kahn in "The Codebreakers" (1967) wrote about cryptanalysis of the Hill Cipher (page 408). He mentions "octogram frequencies" which implies that 8x8 matrices were considered.

*"In general, the Hill system defends itself well against the direct onslaughts of cryptanalysis. Without a knowledge of the basic letter-to-number conversion alphabet, the cryptanalyst may not even be able to start. Even with it, a straightforward frequency-analysis attack is out of the question: octogram frequencies, for example, are hard to collect and even harder to differentiate. Probable words require tedious testing for possible locations and then much mathematical juggling to determine the correct equations; even so, only the relatively trivial trigraphic encipherments have been solved. The cipher has, however, at least one curious chink in its armor. If a cryptanalyst obtains two ciphertexts resulting from a single plaintext enciphered with different involutory equations (of the same type and polygram size), and if he knows the conversion alphabet, he can, in general, recover the equations fairly easily."*

*The real obstacle to practical use of the Hill system is, of course, its ponderousness. Hill sought to minimize this by patenting a device that will encipher small polygrams (up to hexagrams). It consists of a series of geared wheels connected by a sprocketed chain so that the rotation of one wheel will turn all the others, but the range of its keys appears to be limited. Mechanisms could also be built to compute the encipherments of large polygrams, which give the best security, but they would be so complicated that they could not compete on a practical basis with simpler, though possibly less secure, cipher machines. For such reasons, the Hill system has served as a U.S. governmental cryptosystem in only one minor capacity – to encipher the three-letter groups of radio call-signs."*

### Propinquity

Bauer is well-connected in the historical cryptography world. Most communication is non-verbal and people with a close connection to Scheidt and Sanborn have a much better chance of solution than people without such a connection. (Remember the story about ["Masquerade"](#)).

Gilligly mentioned Scheidt's interest in key escrow *"Given Scheidt's interest in key escrow, I'd expect the key to be hidden in there for somebody who knows what to look for"* in 1999 at a time when very few people would know who Scheidt was.

In 2013 he also mentioned *"Scheidt's interest in duress ciphers multi-layered ciphers for which you can give up a key to the enemy that produces credible plaintext without giving up the farm"* which Scheidt mentions in a [2015 video](#).

Also, similarly, people who go to ACA conferences, Kryptos dinners, people with a connection to Washington DC, or the USA in general may have more chance with Kryptos K4 than others not so well connected. Bauer himself works in the adjacent state of Pennsylvania: as with Sanborn he is within 100 miles of the sculpture.

## The last part

["It's in English, plain English"](#) ... and between 95 and 100 characters. Given average word lengths in English, this is expected to be something between 18 and 23 words.

## Which clock?

["There are several really interesting clocks in Berlin"](#) said Sanborn in 2014.

The most obvious candidates are ...

- The Berlin Clock, or Berlinuhr, or Set Theory Clock, or Mengenlehreuhr by Dieter Binniger (1975). What is meant by Sanborn's quote *"you'd better delve into that particular clock"* is anyone's guess. Is it a suggestion for a probable word "THE" or "PARTICULAR" or "THAT" before "BERLINCLOCK"? Is it about a probable word concerning its pre-1995 location at Kurfürstendamm on the corner with Uhlandstraße, or the sculptor Dieter Binniger? Does it suggest a technique involving modulo 11 or modulo 24 arithmetic, or putting the text in a width 11 or width 24 matrix?
- Water Clock, or Clock of Flowing Time, or Wasseruhr, or Uhr der fließenden Zeit by Bernard Gitton (1982) also mentioned in [the NYT article](#).
- World Time Clock (Weltzeituhr) or Atomic Clock (Urania) by Erich John (1969)
- Peace Clock (Friedensuhr) by Jens Lorenz (1989). It has a great [story associated with its unveiling](#). It's been mentioned at [Independent Café](#).
- Rotes Rathaus, the German town hall clock
- A sundial in the Egyptian Museum in Berlin - *"... the results of Borchardt's investigation of the Berlin Clock for an assumed latitude of 25.5°"* in ["Ancient Egyptian Science, A Source Book, Volume Two: Calendars, Clocks and Astronomy"](#) by Marshall Clagett, 1995. The book also examines star clocks, water clocks, shadow clocks, and sundials. This would be a nice way to connect all the parts thematically - the Morse code and K1 concerning shadows, K2 concerning latitude, and K3 concerning ancient Egypt. And then there is Kryptos itself as a sundial, which perhaps suggests "gnomon" as a probable word for the text.

A [German blog](#) of 2017 describes some of the 430 public clocks in Berlin.

Professor Michael J. Sauter has two articles on Berlin clocks historically: [Clock Watchers and Stargazers \(2006\)](#) and [a similar article in 2007](#). The focus is on the Berlin Academy Clock on Unter den Linden.

Also a bit less likely, with no direct connection to Berlin ...

- [The Danish Clock Cryptograph](#) mentioned by [Jose](#)
- [The Polish clock of cryptography for Enigma cryptanalysis](#) mentioned in the Wired article of 2014

## Yahoo group

I've never joined it, but some participants and observers have commented about it online. Unfortunately, the signal to noise ratio seems quite low.

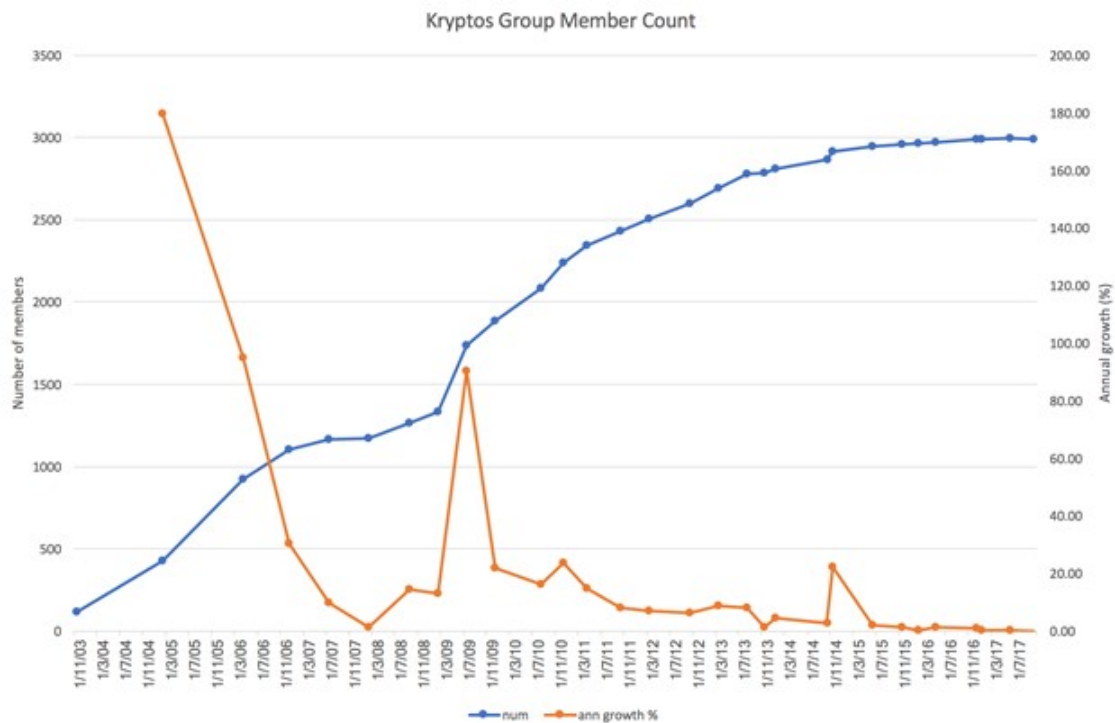
*"So far as I have seen, the so-called Kryptos discussion group hasn't produced anything notable. 'If' the production of "LAYERTWO" using a slightly broken key had been recognized at the time as an indication of an error in the ciphertext, it would be worth noting in the article. However, its significance wasn't appreciated until Sanford [sic] announced the error."* (Wikipedia talk page)

*"The Yahoo group was teeming until Yahoo had the huge hack [gradually disclosed between September 2016 and October 2017]. Then a lot of people just didn't come back."* (reddit)

*"There's a yahoo group for it, too, with an archive of 20,000 messages. Every now and then there is a useful post, but a lot are people who are sure they found the solution but won't share or won't take advice, for a while tons of claims Nostradamus both predicted and solved Kryptos, cranks who are "anagramming" "solutions" (playing alphabet soup)..."* (reddit and cryptocrap)

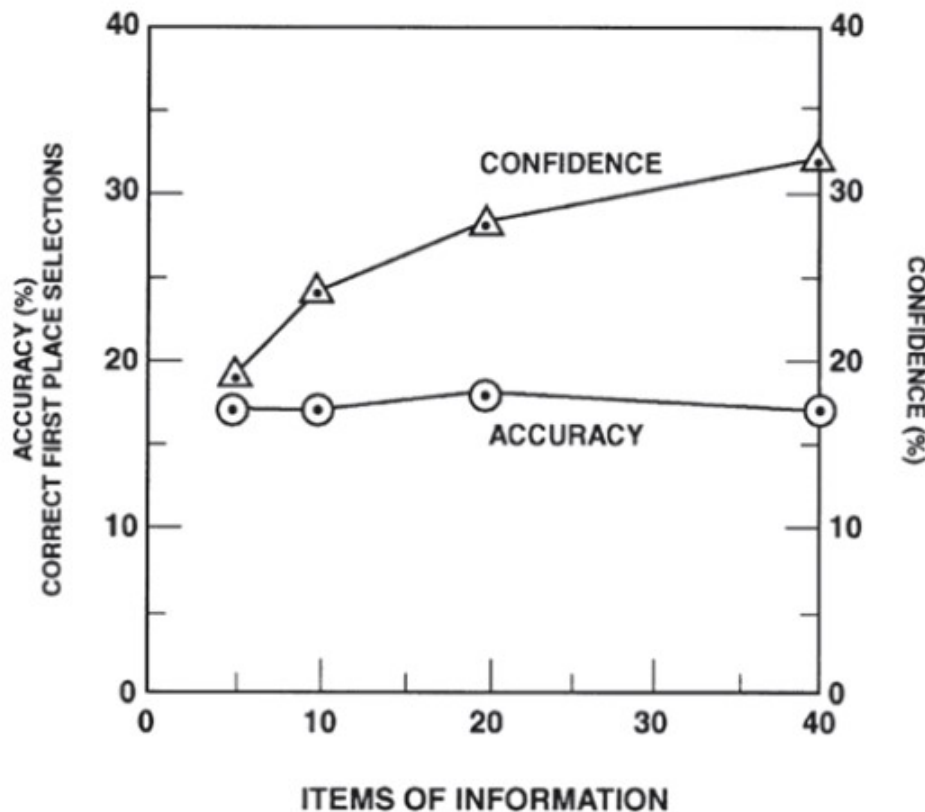
*"The forum seemed to be very active leading up to a conference, but then nothing. However there were like 2 useful posts out of 400."* (reddit)

The membership experienced some small spikes after the November 2010 and November 2014 hints were released but has really flattened off. When the hints were released, or a biennial Kryptos dinner was coming up (2013, 2015, 2017...) there was a flurry of message activity which quickly dropped off again.



Also consider the Figure from [Heuer](#) from the chapter "Do You Really Need More Information?". This graph showed the result of a horserace handicapping experiment (Paul Slovic, "Behavioral problems of adhering to a decision policy", 1973). Having more information can increase confidence but not accuracy.

Figure 5



#### Jose suggestion

This was a nifty idea between November 2010 to November 2014 - a 97 character [suggested plaintext](#) from Reagan's "Tear Down This Wall" speech, with "BERLIN" in characters 64-74.

Jose [plastered it all over comments sections](#) everywhere for years without providing a method. His suggestion has 23 English words. Apart from not providing a method, the other flaws were that it went through a sentence break in the original and started in the middle of a sentence: "... people, to create a safer, freer world. And surely there is no better place than Berlin, the meeting place of East and West".

#### Outlandish theories

As Schridde has said, the theories of running key, auto key, and combined transposition and polyalphabetic would have to be regarded as unlikely after the "BERLIN CLOCK" clues. Other suggestions are very difficult systems to analyse given the short cipher text length. Enigma requires a knowledge of rotor settings and the classical version doesn't allow letters to encrypt to themselves (as with "MZFPK" to "CLOCK").

- Enigma. ["An Enigma that short would be quite a trick"](#) (Gillooly). The standard Enigma machine wouldn't map a "K" to itself in the ciphertext.
- M-94 and M-209. Gillooly mentioned ["The only likely periodicity appears to be at period 25, but that may well just be chance"](#). Does this indicate a particular likely system associated with the number 25, like [M-94](#)? He also [wrote](#) *"I don't see any regularities other than a probably-spurious Phillips-like distribution, if you ignore the presence of both I and J."*
- Fractionated Morse. Possible, but very difficult...

- Hill Cipher with a different number of characters e.g. 27 or 29
- Vigenere followed by Hill Cipher
- Transposition followed by Hill Cipher - but then saying "NYPVTT" maps to "BERLIN" becomes difficult
- Gromark – to help explain the high index of coincidence after arranging the text at width 50. See [Gromark subpage](#).
- Trifid cipher – since Scheidt commented there were 98 characters remaining, we could include “?” as an extra character, which gives us 27 different characters in the ciphertext as a Trifid cipher would have. I haven’t looked at this as I assume this has been done to death already, although I’m aware the keyspace for the encryption alphabet is huge with 27! possibilities. Update – I created [another page about this idea](#).
- One time pad (as suggested by the initial [CIA memo](#) - in which case you'd have to find a key which had been used elsewhere... and then it wouldn't be a real OTP ... like [Venona](#))

Curiously, a Washington Post article of January 1990 about Kryptos hasn’t been transcribed online. The relevant part (from the [NSA presentation](#)) states

*The secret phrase will be cut into the plate. Anyone who knows a coding system called the Vigenere Tableau, invented in 1586 by French diplomat Blaise de Vigenere, will be able to decipher one-half of the phrase. The other half will be encoded in **a modern system created for the project** by an expert cryptographer, whom Sanborn would not identify.*

#### Other statistical analysis

For information about the period 7 properties, see [the Schridde page](#).

The [original NSA document](#) stated:

*A statistical analysis of this portion showed some roughness on interval 7. This could be a characteristic of plain-text auto-key, if the alphabet used has a high frequency letter assigned the value of 0. Another hypothesis is that this last section employs both of the systems already used. First the message is encrypted using some set of alphabets, as was done in the first and third breakthroughs, and then the cipher is put through transposition, such as that used in the second breakthrough. If the original text had a repeat at a distance of 7 apart (or perhaps 14 or even 21 apart), then after transposing the text, the repeat would now show up in the interval statistic rather than the width statistic.*

Back in 1999 Doug Gwyn [looked at ICs of the delta text](#) - although with the 2010 clue that there's a one-to-one correspondence between "BERLIN" and "NYPVTT" this would have to be considered a defunct possibility.

[Monet's Kryptos observations](#) (last updated February 2014) is more observations than solution attempts.

#### Western Union payment for Sanborn to examine solutions

Western Union isn't available in Iran, Myanmar, Somalia or North Korea so post 2014, people from those countries have to try questioning Jim Sanborn another way. The [\\$50 payment](#) he requests is really for crank deterrence. It may also be deterring interest in the sculpture as it could be interpreted as a money-making venture.

#### Google n-gram search

Apart from the huge growth in computer power since 1990, Google has also been busy digitising all the books in the world. They have a [page of n-gram data](#) (updated 2012) so that if you want to download all the 2-grams, you can see what words tend to come before or after "BERLIN" or "CLOCK" in English books, and in 3-grams, maybe you can find the same for "BERLIN CLOCK" if it's a popular enough phrase.

#### Total immersion

People who become totally immersed in cryptanalysis sometimes have nervous breakdowns ([William F. Friedman](#)) or engage in wishful thinking about famous unsolved cipher solutions (Beale, Voynich, Zodiac).

Ideally a cryptanalysis task would be a positive and growth experience, not a complete waste of time, and would help each individual involved to be a more well-rounded person. (Although I can well imagine that working on Venona was incredibly boring).

Charles Babbage wrote “*I am myself inclined to think that deciphering is an affair of time, ingenuity, and patience; and that very few*

*ciphers are worth the trouble of unravelling them."*

Parker Hitt wrote *"Success in dealing with unknown ciphers is measured by these four things in the order named: perseverance, careful methods of analysis, intuition, luck. The ability at least to read the language of the original text is very desirable but not essential."*

Mark Siegal in The Magazine stated *"Kryptos has motivated me to do more enthusiastic programming than almost anything else in my life."*

*"This may be a good cryptosystem or it may not, but if it takes much effort to crack, people are only going to attempt it if they have sufficient motivation."* – Doug Gwyn, net.crypt, 1985

*"My guess is that when it is finally cracked, it will be by somebody versed in classical C/A who happens to make a lucky guess (about the key, the general system, or ???) that enables further progress. That actually happens a lot in cryptanalysis and is part of the "art"."* – Doug Gwyn, sci.crypt, 2004

[Sanborn commented](#) ... *"... there are actually schools for the disabled and autistic children, who have used Kryptos as therapy. I think it's a wonderful thing. In addition there are - on the opposite side - there are people whose relationships have been destroyed over Kryptos."*

This was followed by another Sanborn style exchange about what he really meant by following Kryptos around the country, reminiscent of the bizarre ["Something you did do?" "Something I could have done"](#) exchange in Wired. (I'm missing the non-verbal context, but I'd find someone who talked like that quite annoying.)

This in turn reminded me of his [2005 throwaway line](#) in Wired which probably discouraged many: *"If a person deciphers and sends me the exact decipherment -- if it can be deciphered exactly, considering most of my things are rife with mistakes on purpose -- I'd probably let them know that they got it if they did."* Or, reading that more charitably, is that supposed to mean there could be more than one solution to K4, like a duress cipher?

The idea of using a Hill cipher with non-invertible matrices was examined by Bauer et al, and briefly in one sentence at [PC's Xcetra Support](#) page, although the vast majority of that page is deep in "wishful thinking" territory. Hill Ciphers with different sized blocks [were mentioned on theoryland](#) but you could get the plaintext to be anything that way.

Lambros D. Callimahos's [NSA classes](#) certainly seemed to encourage becoming well-rounded, with trivia questions long before the age of Google.

[Kryptosfan's blog](#) has a big list of movies and books that he's read. I've picked up a few books that I probably wouldn't otherwise have read, and have been working my way through. For example:

- "A Spy in Rome" by Peter Jenkins - [recommended by Apex](#).
- "The Codebreakers" by David Kahn (1996 update). Fantastic background and very detailed.
- ["The Psychology of Intelligence Analysis"](#) by Richards Heuer, a 1999 book published by the CIA. Reading this book and applying the theories will help you "keep your feet on the ground" and hopefully help you realise when you're engaging in wishful thinking. See also [Pitfalls of decipherment](#).
- "Killing Hope" by William Blum and "Legacy of Ashes" by Tim Weiner, about the history of the CIA
- "Merchants of Menace" by Peter Butt, about the Nugan Hand Bank, a CIA bank of the 70s and 80s - a successor to ["The Crimes of Patriots"](#) by Jonathan Kwitny from 1987
- ["Encyclopedia of the CIA"](#) - Kryptos gets a mention of course
- "Ancient Egyptian Science" mentioned above, though it's quite dry and detailed
- The two Michael Sauter articles mentioned above
- "Unsolved" by Craig Bauer and "The Mathematics of Secrets" by Joshua Holden

(And whatever happened to the book about Kryptos Sanborn talked about writing [in 2009](#) and [2010?](#))

### Similar challenges

The idea of a public puzzle which has remained unsolved for several years has been seen before in books.

*Masquerade* by Kit Williams, published August 1979, solved March 1982

*The Code Book* by Simon Singh, published September 1999, solved October 2000

*Can you crack the Enigma Code* by Paul Belfield, published September 2006, solved [August 2012](#)

These UK books attract the same kind of people as Kryptos which is why there are Kryptos pages on the [Quest4Treasure forum](#) and the [Tweleve forum](#). One difference between these books and Kryptos is that it is certain the book puzzles were rigorously externally verified. We are essentially trusting Scheidt that K4 is decipherable.

#### Extra hints

So based on revelations in 2006, 2010, and 2014 I guess it would be nice to have arithmetic progression and get a new clue in 2018. Ideally, it would be a hint about the method, like the zetaboards comments above said, but a list of the word divisions – like turning a “Patristocrat” into an “Aristocrat” - or an extra word might also help.

In comparison to the [Code Book Challenge](#) everything here is as clear as mud.

#### Last words

From the sculptor (2012):

*"Like Kryptos, the other public works are designed to exude their information slowly. ... For the past 30 years, my task as an artist has been to release this hidden information at a rate commensurate with its importance, and at the time of my choosing so as to prolong the experience of discovery. As we all know, artwork that gives up its form or content quickly is soon forgotten."*

Given all the above, it would be wise not to spend too much time on this and wait for some new information to be disclosed.